

2020 AMC 10B Problem 13 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 13. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 13

Problem:

Andy the Ant lives on a coordinate plane and is currently at $(-20, 20)$ facing east (that is, in the positive x -direction). Andy moves 1 unit and then turns 90° left. From there, Andy moves 2 units (north) and then turns 90° left. He then moves 3 units (west) and again turns 90° left. Andy continues this process, increasing his distance each time by 1 unit and always turning left. What is the location of the point at which Andy makes the 2020th left turn?

Answer Choices:

- A. $(-1030, -994)$
- B. $(-1030, -990)$
- C. $(-1026, -994)$
- D. $(-1026, -990)$
- E. $(-1022, -994)$

Solution:

Andy makes a total of 2020 moves: 1010 horizontal (505 left and 505 right) and 1010 vertical (505 up and 505 down). The x -coordinate of Andy's final position is

$$-20 + \overbrace{\underbrace{1-3}_{-2} + \underbrace{5-7}_{-2} + \underbrace{9-11}_{-2} + \cdots + \underbrace{2017-2019}_{-2}}^{1010 \text{ terms, } 505 \text{ pairs}} = -20 - 2.505 = -1030$$

The y -coordinate of Andy's final position is

$$20 + \overbrace{\underbrace{2-4}_{-2} + \underbrace{6-8}_{-2} + \underbrace{10-12}_{-2} + \cdots + \underbrace{2018-2020}_{-2}}^{1010 \text{ terms, } 505 \text{ pairs}} = 20 - 2 \cdot 505 = -990$$

Together, we have $(x, y) = (\text{B}) \boxed{(-1030, -990)}$.

OR

You can find that every four moves both coordinates decrease by 2. Therefore, both coordinates need to decrease by two 505 times. You subtract, giving you the answer of $(\text{B}) \boxed{(-1030, -990)}$.

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